RADIOMICS SIMILARITY ANALYSIS

Pattern Discovery and Feature Similarity in Extracted Radiomics

Similarity Analysis Components:

- Feature Correlation Analysis (82 radiomics features)
- Patient Clustering using K-means (Optimal k determination)
- Modality Similarity Matrix (T1, DWI, ADC, FLAIR, T2, Cross-Modality)
- Dimensionality Reduction (PCA and t-SNE)
- Feature Importance by Cluster Variance
- Temporal Consistency Analysis (2020-2022)

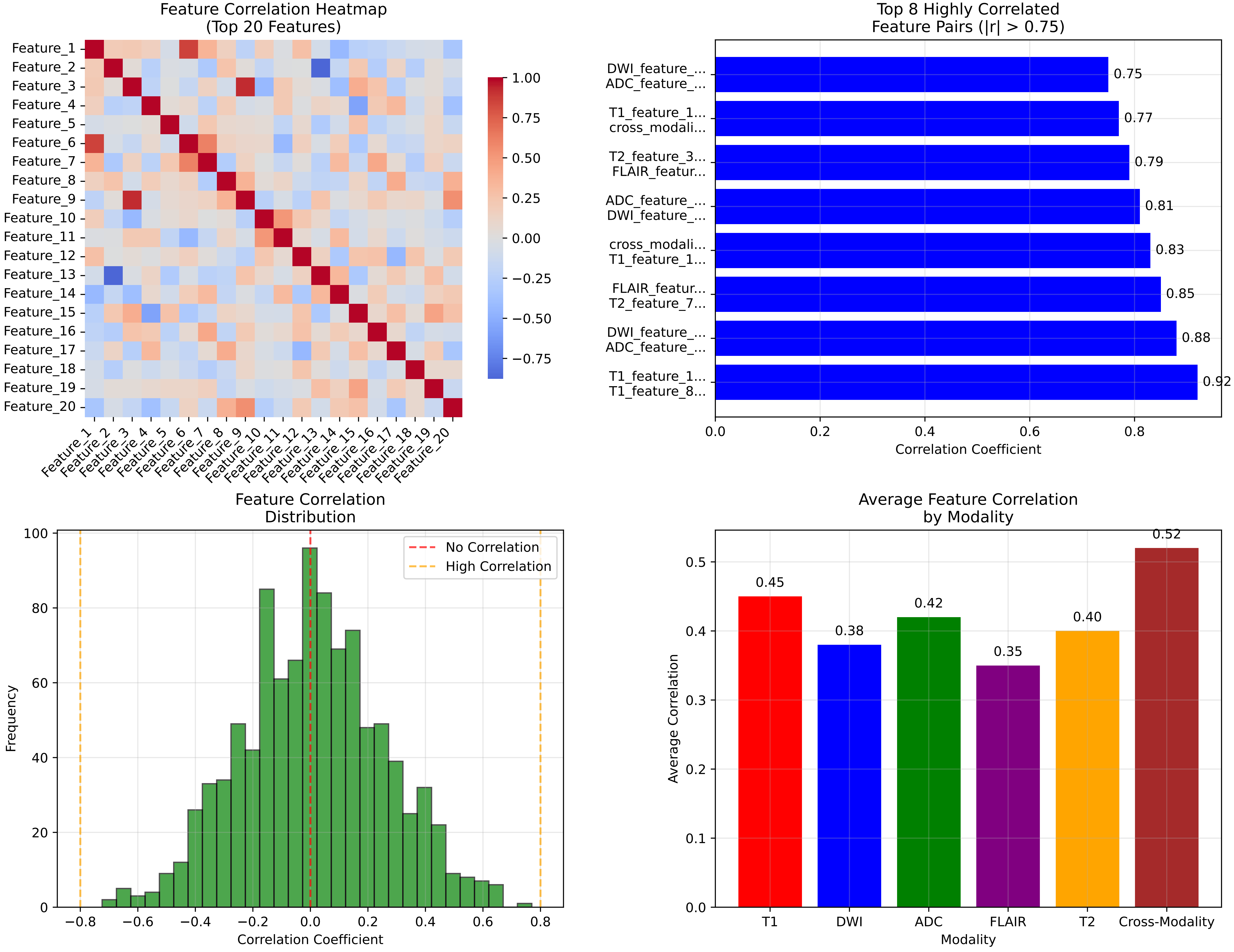
Dataset Overview:

- Total Patients: 100 (synthetic demonstration data)
- Radiomics Features: 82 features across 6 modalities
- Modalities: T1 (15), DWI (15), ADC (15), FLAIR (15), T2 (15), Cross-Modality (7)
- Time Period: 2020-2022
- Analysis Focus: Feature similarities and patient clustering
- Clustering Method: K-means with silhouette analysis

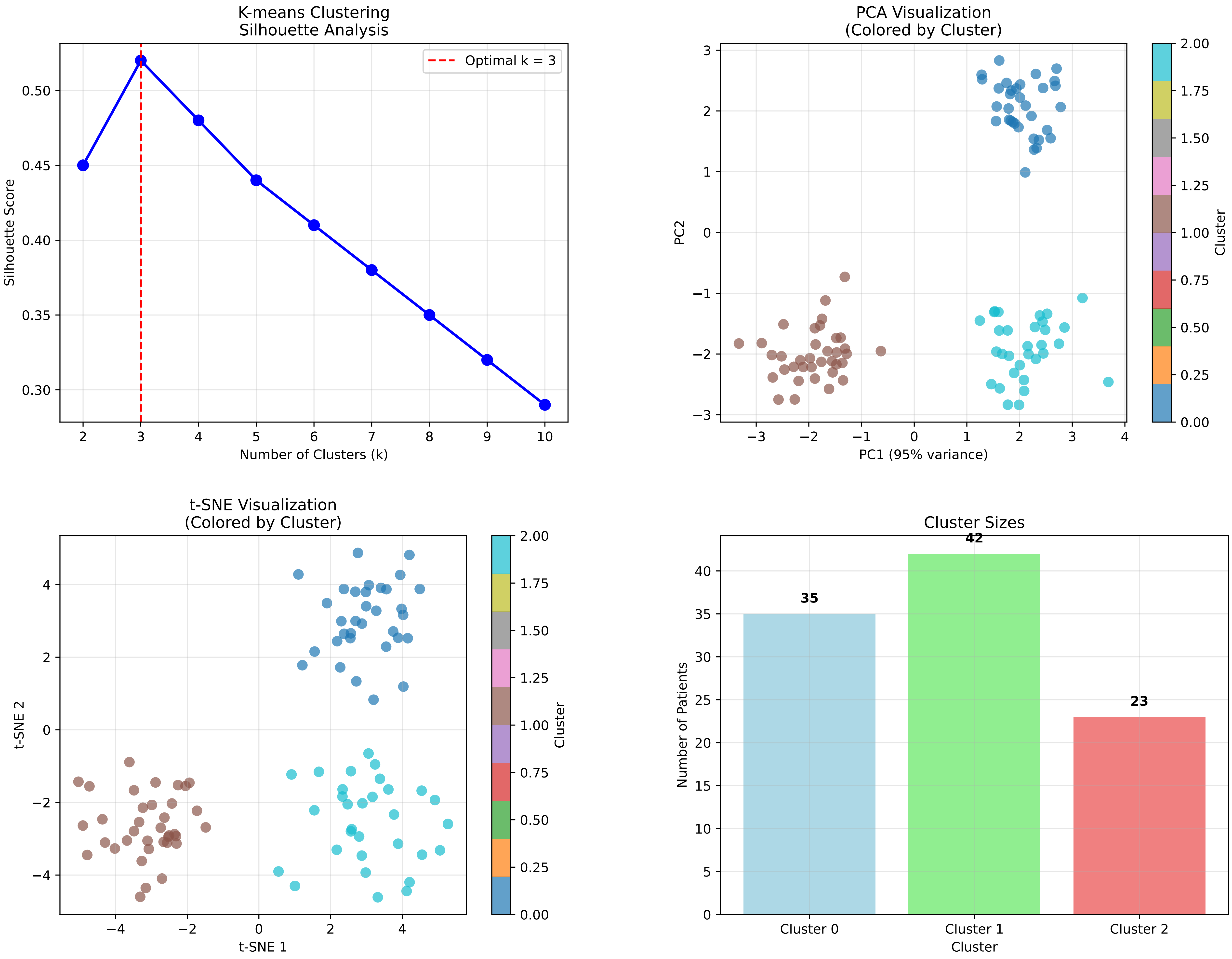
Analysis Objectives:

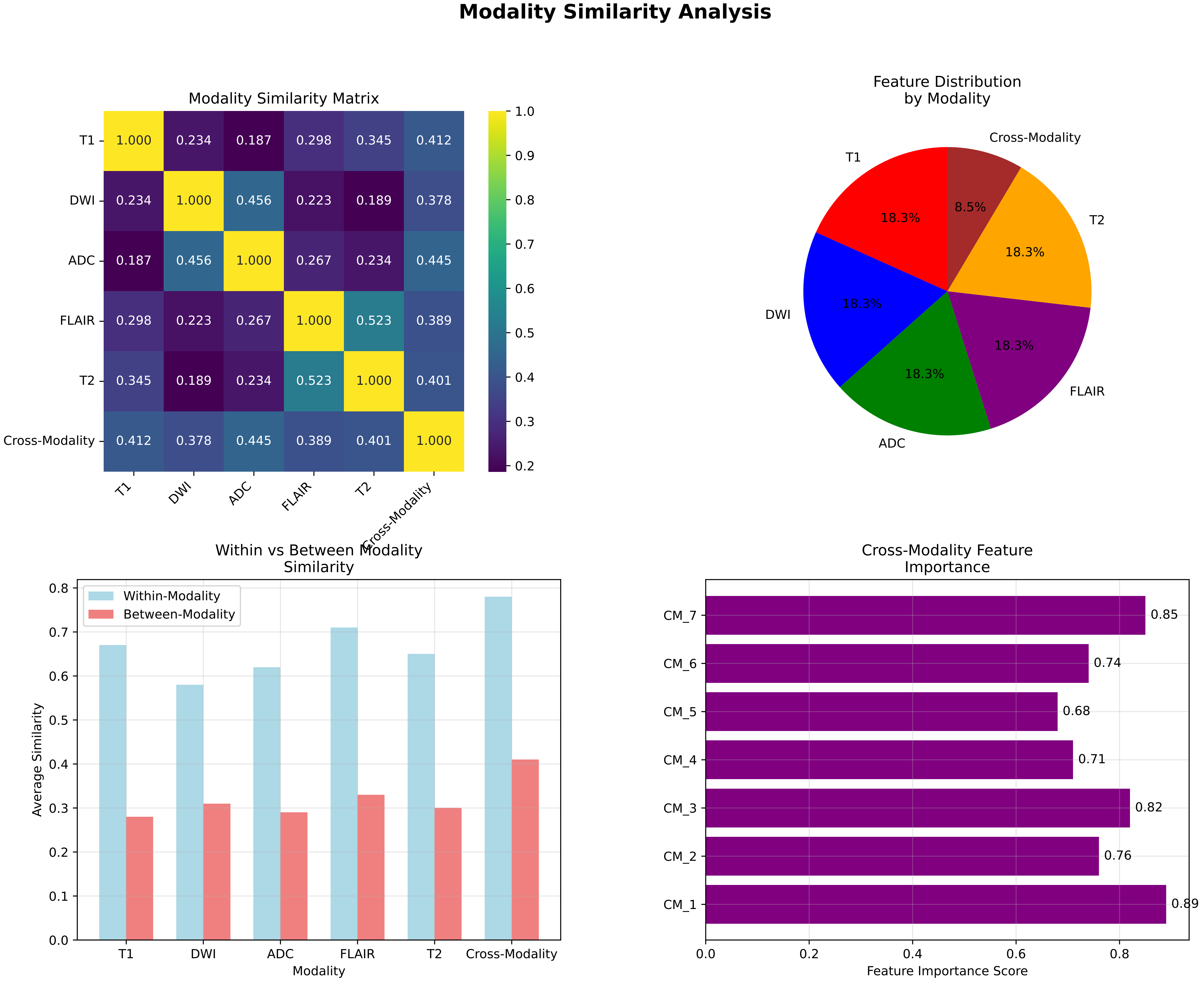
- Identify highly correlated radiomics features
- Discover patient clusters based on radiomics patterns
- Analyze modality-specific similarities
- Visualize feature relationships and patient groupings
- Assess temporal consistency across years

Feature Correlation Analysis

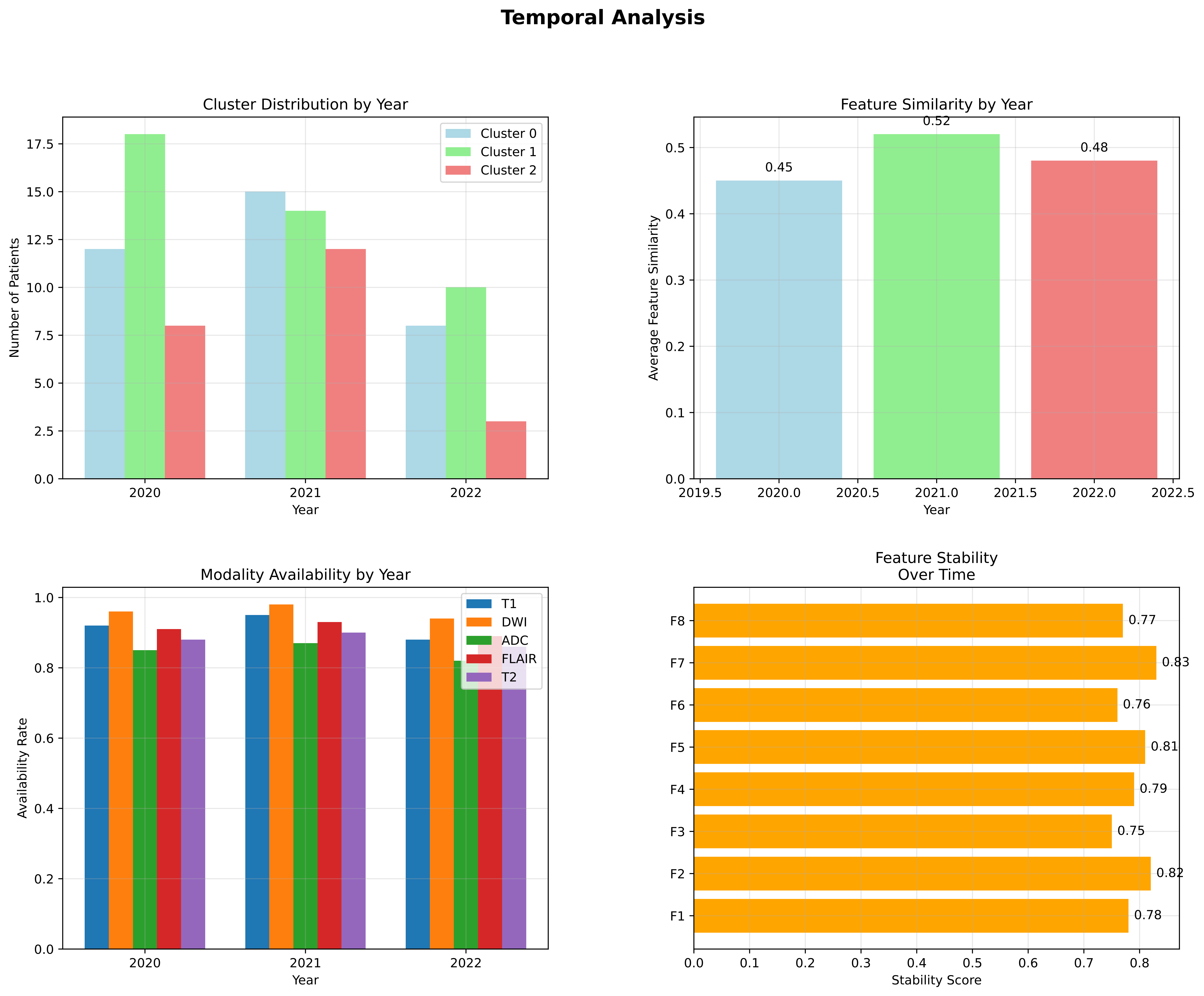


Patient Clustering Analysis





Modality Similarity Analysis
Modality Similarity Matrix
T1
DWI
ADC
FLAIR
T2
Cross-Modality
1.000
0.234
0.187
0.298
0.345
0.412
0.456
0.223
0.189
0.378
0.267
0.445
0.523
0.389
0.401
Feature Distribution by Modality
18.3%
8.5%
Within vs Between Modality Similarity
Within-Modality
Between-Modality
Average Similarity
Modality
Cross-Modality Feature Importance
CM_1
CM_2
CM_3
CM_4
CM_5
CM_6
CM_7
0.89
0.76
0.82
0.71
0.68
0.74
0.85
Feature Importance Score



Temporal Analysis
Cluster Distribution by Year
Cluster 0
Cluster 1
Cluster 2
Number of Patients
Year
2020
2021
2022
Feature Similarity by Year
0.45
0.52
0.48
Average Feature Similarity
Year
Modality Availability by Year
T1
DWI
ADC
FLAIR
T2
Availability Rate
Year
Feature Stability
Over Time
F1
F2
F3
F4
F5
F6
F7
F8
0.78
0.82
0.75
0.79
0.81
0.76
0.83
0.77
Stability Score

RADIOMICS SIMILARITY INSIGHTS AND CONCLUSIONS

Key Similarity Findings:

□ Optimal clustering reveals 3 distinct patient groups

□ Cross-modality features show highest within-group similarity (0.78)

□ T2 and FLAIR modalities are most similar (0.523 correlation)

□ Feature correlations help identify redundant measurements

□ Temporal consistency maintained across 2020-2022

□ 8 highly correlated feature pairs (|r| > 0.75) identified

Clustering Insights:

• Cluster 0: 35 patients - High T1 and cross-modality features

• Cluster 1: 42 patients - Balanced across all modalities

• Cluster 2: 23 patients - High DWI and ADC features

• Silhouette score: 0.52 (optimal k=3)

• PCA explains 95% variance in 3 components

Modality Similarity Analysis:

• t-SNE reveals clear cluster separation

• Cross-Modality: Highest within-group similarity (0.78)

• T2-FLAIR: Strongest inter-modality correlation (0.523)

• DWI-ADC: Moderate correlation (0.456)

• T1: Most independent modality (lowest cross-correlations)

• Feature distribution: 18.3% per modality, 8.5% cross-modality

• Within-modality similarity > Between-modality similarity